MIL-E-1/1595A(NAVY)
1 February 1971
SUPERSEDING
MIL-E-1/1595(NAVY)
26 November 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS

TYPE 8376

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Hydrogen diode

Outline: See figure 1

Mounting position: Horizontal or vertical with base down

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	ib	Ib	Ip	Fault current (0.15 sec max)
Unit:	Vac	kv	a	Adc	Aac	a
Maximum:						
Rectifier: (Note 1)	5.0 ±5%	25	10	2.5	---	90
Clipper Diode:	5.0 ±5%	33	500	1	20	1500
Charging Diode:	5.0 ±5%	33	7.5	2.5	---	90
Backswing Diode:	5.0 ±5%	33	500	2	20	1500
TEST CONDITIONS:	5.0	---	---	---	---	----

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	tk	Eres	TA	TE
Unit:	sec	Vac	°C	°C
Maximum:	(min)			(Note 2)
Rectifier:	180	5.0 ±5%	+125 -50	400
Clipper Diode:	180	5.0 ±5%	+125 -50	400
Charging Diode:	180	5.0 ±5%	+125 -50	400
Backswing Diode:	180	5.0 ±5%	+125 -50	400
TEST CONDITIONS:	---	5.0	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Conditions	AQL (Percent Defective)	Insp. level or code	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 1							
1231	Emission	Ib=3Adc; f=60Hz	0.65	II	etd	---	70	v
1301	Heater current		0.65	II	If	12	15	Aac
1301	Heater current (reservoir)		0.65	II	Ires	2	3	Aac
3249	Rectifier operation	Ef=Eres=5.25Vac; Epx=27.5kv; Ib= 2.2Adc(min) (Notes 5 and 6)	0.65	II	---	---	---	---
3249	Clipper diode operation	(Notes 6 and 7) Epx=33kv; prr=310	0.65	II	---	---	---	---
3265	Hydrogen diode ionization voltage	Ef=Eres=4.75Vac (Notes 7 and 8)	0.65	II	Ez	---	500	v
	Quality conformance inspection, part 2							
1031	Sweep frequency vibration	f=50 to 1000 to 50Hz; g=10; planes X, Y, and Z; t=10 minutes per plane, 30 minutes total	---	(Note 4)	---	---	---	---
---	Post sweep frequency vibration end point	Rectifier operation	---	---	---	---	---	---
1031	Variable frequency vibration	No voltages (Notes 3 and 4) f=5 to 50 to 5Hz	0.65	S3	---	---	---	---
1042	Shock specified pulse	Shock=11±2ms; Acceleration= 50g peak(min); 3 planes; Ef= Eres=5Vac (Notes 9 and 14)	---	(Notes 10 and 11)	---	---	---	---
---	Post variable frequency vibration and shock end point	Rectifier operation	---	---	---	---	---	---
3249	Pulse transformer backswing diode operation	(Notes 6 and 15)	0.65	S3	Ez	---	500	v

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Method	Requirement or test	Conditions	AQL (Percent Defective)	Insp. level or code	Symbol	Limits		Unit
						Min	Max	
	Quality conformance inspection, part 3							
---	Life-test provisions (rectifier)	Group C; (Notes 5 and 12) Operation of gas rectifiers; Epx= 25kv; t=500hrs	---	---				
---	Life-test end points (rectifier)	Ionization voltage	---	---	Ez	---	180	v
---	Life-test provisions (charging diode)	Group C; (Notes 12 and 13) Epx=33kv; PRF= 310pps; t=500hrs	---	---				
---	Life-test end point (charging diode)	Ionization voltage	---	---	Ez	---	500	v
---	Life-test provisions (clipper diode)	Group C; (Notes 7 and 12) Epx=33kv; PRF= 310pps; t=500hrs	---	---	---	---	---	---
---	Life-test end points (clipper diode)	Hydrogen diode ionization voltage	---	---	Ez	---	750	v

NOTES:

1. Maximum current and voltage ratings do not apply simultaneously. See figure 3 for current versus voltage readings.

2. No external portion of the tube shall be permitted to exceed this maximum temperature.

3. Total time of vibration in each plane shall be 10 minutes.

4. This test shall be performed on a minimum of two tubes.

5. The following circuit constants shall apply: Epx=25kv, Ib=2 Adc; and ib=8a. The minimum current passing through the two tubes shall be not less than 90 amperes for 0.1 second should an arc-back of either tube occur. One-half the specified 27.5kv inverse voltage shall be applied initially; voltage shall be increased gradually to full value in approximately one minute.

6. The tube shall go into conduction smoothly and shall operate for 30 minutes with one arc-back permitted in the first 15 minutes and none thereafter.

7. The tube shall be operated as a line clipper in a thyratron modulator shown on figure 3249-2 with the following circuit constants: peak thyratron forward voltage = 33kv, R_L adjusted for line clipper, ib=500a(min), tp=2.5±0.25us, resonant charging at prr=500pps.

8. The tube shall be operated as shown on figure 3249-2 (circuit for operation of line clipper diodes).

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9. The tubes shall be subjected to two long duration shocks in the direction of each of the three mutually perpendicular axes (X, Y, Z) for a total of six blows. This is a non-destructive test. The following is a destructive test, and it shall be performed once in each 18 month interval: The tube shall be subjected to seven long duration shocks in the direction of each of the three mutually perpendicular axes (X, Y, Z) for a total of 21 blows.

10. Shock test sampling and acceptance criteria shall be as follows: For one month's production to be sampled up to 250 units, the first sample (n1) shall consist of one tube with an acceptance number (c1) of zero. Second sample (n2) shall consist of two tubes with an acceptance number (c2) equal to 1 for the combined sample.

11. This test shall be performed on a minimum of three tubes for qualification.

12. Life test shall be shut down every 96 hours for a 60-minute interval.

13. The tube shall be tested as a charging diode in the test circuit shown on figure 3249-4. Constants: peak thyratron forward voltage = 33kv, Ib=1.5 amperes (min), charging circuit resonance = 500pps.

14. The tube shall be monitored for electrical discontinuities during the shock test.

15. The tube shall be tested in the test circuit shown on figure 3249-3. The circuit constants shall be chosen so that epx = 16kv (min), ib = 500. The current pulse through the tube shall be 3us, the repetition test frequency, 310pps.

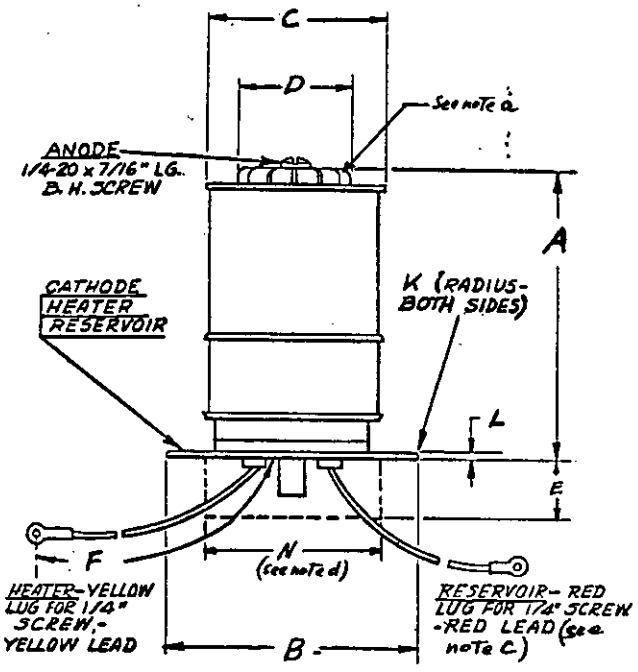
User activities:

Navy - AS, OS, MC, CG, SH

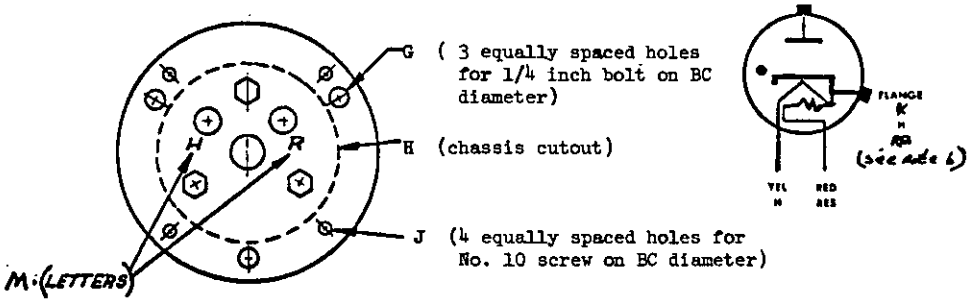
Preparing activity:
Navy - EC

(Project 5960-N887)

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L_{T_R}	DIMENSIONS	
	Minimum	Maximum
QC 1, Part 2		
A	5.000(127.00)	5.129(130.15)
B	4.370(111.00)	4.380(111.25)
C		3.06(77.72)
D	1.975(50.17)	2.025(51.44)
E	1.00(25.4	
F	10.50(241.3)	11.00(254.0)
G	3.730(94.74)	3.770(95.76)
H	2.59(65.79)	
J	3.933(99.90)	3.973(100.91)
Reference dimensions		
K	.063(1.60)	
L	.125(3.18)	
M	.188(4.78)	
N	3.07(77.98)	



NOTES:

a. The use of an anode radiator is optional. If one is used it shall conform to the outline as shown.

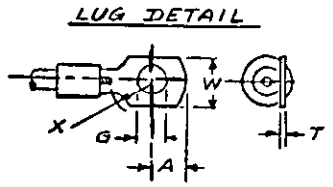
b. In order to avoid damage to tube, the cathode connection must be made to the base flange only.

c. Lead terminal shall be in accordance with figure 2. Periodic-check inspection shall apply.

d. Below chassis clearance.

FIGURE 1. Outline drawing of election tube type 8376.

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T	0.040 MIN.
G	0.260-0.313
W	0.460-0.480
A	0.305 MAX.
X	0.380 MIN.

FIGURE 2. Lug terminal details.

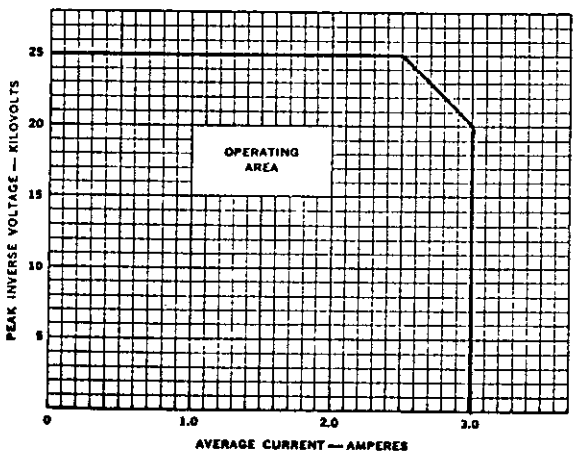


FIGURE 3. Current versus voltage chart.

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